# Polynomial Regression Report

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## 1 Introduction

This report documents the method, implementation, and justification for building high degree polynomial regression models to solve two real world predictive tasks. The tasks require learning complex non linear relationships from limited sensor data. Due to the high risk of overfitting associated with high degree polynomials, this project adopts a rigorous machine learning pipeline encompassing standardisation, 10 fold cross-validation, and penalised regression techniques (Ridge, Lasso, and Elastic Net).

The two scenarios modelled are:

1. **Phase 1 (var1):** Predicting plant power output using 6 sensory variables (up to degree 10).
2. **Phase 2 (var2):** Subterranean Thermal Reservoir Mapping based on 3D spatial coordinates (up to degree 20).

## 2 Method

### 2.1 Polynomial Expansion

To capture the highly complex geological and thermodynamic structures inherent in the data, the base features $x \in \mathbb{R}^n$ were mapped into a high-dimensional polynomial space. A polynomial of degree $d$ includes all monomial terms such that the sum of the powers of the features is at most $d$.

For Phase 1 ($n_{\text{features}} = 6, d = 10$), this results in 8,007 features. For Phase 2 ($n_{\text{features}} = 3, d = 20$), this results in 1,770 features. Let the design matrix of these engineered features be denoted as $X \in \mathbb{R}^{N \times p}$, where $N$ is the number of samples.

### 2.2 Feature Standardisation

When engineering polynomial features, the numerical scales of the columns diverge exponentially. For instance, if an input variable is 0.5, its 10th-degree term becomes $0.5^{10} \approx 0.00097$.

If the design matrix is not scaled, penalised regression algorithms (which penalise the magnitude of parameters) will unfairly penalise the coefficients of larger features while ignoring tiny features. Therefore, before any model fitting, the design matrix $X$ is strictly standardised such that every column has zero mean and unit variance. To prevent data leakage, the standardisation parameters (mean and standard deviation) are computed strictly on the training folds and applied uniformly to validation and test data.

### 2.3 Regularisation Algorithms

Given that the number of polynomial features $p$ frequently exceeds the number of available data points $N$, standard Least Squares is highly susceptible to severe overfitting and singular matrices. To counteract this, three regularisation strategies were employed:

#### 2.3.1 Ridge Regression ($L_2$ Penalty)

Ridge regression shrinks coefficients uniformly by penalising the squared $L_2$-norm of the parameters $\beta$. The objective function is:

$$J_{\text{Ridge}}(\beta) = \frac{1}{N}\|X\beta - y\|_2^2 + \lambda\|\beta\|_2^2 \tag{1}$$

#### 2.3.2 Lasso Regression ($L_1$ Penalty)

Lasso applies an $L_1$ penalty, forcing the model to perform feature selection by shrinking less important polynomial coefficients to exactly zero.

$$J_{\text{Lasso}}(\beta) = \frac{1}{N}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1 \tag{2}$$

This is theoretically crucial for Phase 1 (degree 10), as it is highly probable that only a sparse subset of the 8,007 features represents the true underlying signal.

#### 2.3.3 Elastic Net

Elastic Net provides a middle ground, linearly combining $L_1$ and $L_2$ penalties. It retains the feature selection property of Lasso while maintaining the stability of Ridge when dealing with collinear features.

$$\hat{\beta} = \arg\min_{\beta} \left(\|y - X\beta\|_2^2 + \lambda_2\|\beta\|_2^2 + \lambda_1\|\beta\|_1\right) \tag{3}$$

Where $\rho$ is the L1-ratio.

# 3 Model Training and Selection Pipeline

## 3.1 Data Splitting and The Hold-out Set

To ensure the reported metrics are completely unbiased, the provided training data (1,000 points) was split. 20% (200 points) was completely locked away as a "hold-out" set. The remaining 80% (800 points) served as the primary development set for model selection.

## 3.2 $K$ Fold Cross-Validation

For model selection, a 10 fold cross-validation strategy was applied to the 800 development data points. For every possible combination of:

- Polynomial Degree (1 to 10 for Phase 1; 1 to 20 for Phase 2)
- Model Family (Normal Least Squares, Ridge, Lasso, Elastic Net)
- Hyper-parameter ($\lambda, \alpha, \rho$ values)

The model was trained on 9 folds and evaluated on the 1 left-out fold. This process was repeated 10 times, and the Mean Squared Error (MSE) was averaged across all folds.

# 4 Results and Rationale

## 4.1 Evaluation Metrics

Models were evaluated using two primary metrics:

1. **Mean Squared Error (MSE):** Measures the average squared difference between predictions and actuals. This served as the primary loss function during cross-validation.
2. $R^2$ **Score (Coefficient of Determination):** Evaluates the proportion of variance in the target variable captured by the polynomial model.

## 4.2 Phase 1: Factory Power Optimisation (var1)

After exhaustively evaluating all degrees up to 10 across Normal Least Squares, Ridge, Lasso, and ElasticNet, the 10-fold cross-validation identified that the optimal architecture was an **Elastic Net** model of degree **5**, with regularisation parameters $\alpha \approx \mathbf{0.00686}$ and $L_1$-ratio $= \mathbf{0.95}$.

**Rationale:** The selection of Elastic Net highlights the underlying nature of the data. At degree 5, the model expanded 6 variables into 461 polynomial features. The overwhelming preference for an $L_1$-heavy penalty ($L_1$-ratio $= 0.95$ makes it nearly identical to pure Lasso) conclusively demonstrates that the true physical function driving factory

power is extremely sparse. The model successfully discarded hundreds of noise-inducing polynomial interactions by driving their coefficients to exactly zero, keeping only the most critical features.

On the untouched 20% hold-out set, this optimal architecture achieved an unbiased **MSE of 0.3566** and an $R^2$ **of 0.9705**.

### 4.3 Phase 2: Subterranean Thermal Mapping (var2)

The optimal cross-validated architecture was a **Ridge Regression** model of degree **10** with a regularisation penalty of $\lambda \approx \mathbf{0.811}$.

**Rationale:** The heat map represented a highly complex non-linear surface. Because subterranean thermal propagation is continuous and diffuses smoothly in 3D space, the underlying function is extremely dense, not sparse. Consequently, Ridge Regression outperformed Lasso and Elastic Net. The $L_2$ penalty successfully prevented the model from overfitting to the specific noise in the physical core samples, while allowing it to leverage deep non-linearities up to degree 10 by shrinking correlated polynomial coefficients smoothly together.

On the untouched 20% hold-out set, the Phase 2 model achieved an unbiased **MSE of 0.2576** and an $R^2$ **of 0.9948**.

## 5 Conclusion

By designing a pipeline that strictly segregates training data, dynamically scales features to prevent unequal penalisation, and cross-validates across both model architectures and regularisation hyper-parameters, we successfully generated highly accurate, robust predictions. Finally, before predicting the blind test sets, the optimal architectures were retrained on the entire 1,000 point datasets to ensure the final weights were as accurate as theoretically possible.